

AWS re:Invent

NOV. 28 – DEC. 2, 2022 | LAS VEGAS, NV

D O P 3 0 5

Creating custom constructs with AWS CDK

Natalie White (she/her)

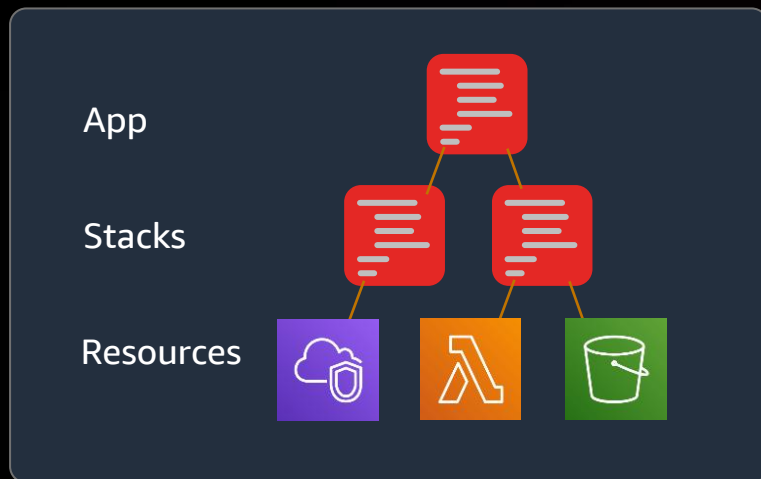
Principal Solutions Architect
AWS

David Christian (he/him)

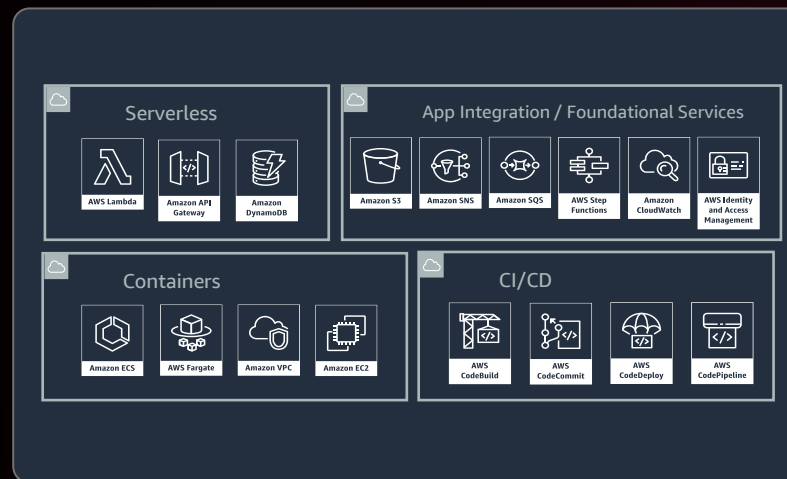
Principal Solutions Architect
AWS



Main components



Core framework



AWS Construct Library

```
cdk-app on p master [?] ls @v0.1.0 via v14.2.0 using anish-cdk took 10s
> cdk diff
Stack CdkAppStack
IAM Statement Changes
+


| Resource              | Effect | Action          | Principal                 | Condition                                                       |
|-----------------------|--------|-----------------|---------------------------|-----------------------------------------------------------------|
| ! \${CdkAppQueue.Arn} | Allow  | sqs:SendMessage | Service:sns.amazonaws.com | {   "ArnEquals": {     "aws:SourceArn": "\${CdkAppTopic}"   } } |


(NOTE: There may be security-related changes not in this list. See https://github.com/aws/aws-cdk/issues/1299)

Conditions
[+] Condition CDKMetadataAvailable: {"Fn::Or":[{"Fn::Or":[{"Fn::Equals":{"Ref":"AWS:Region"},"ap-east-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"ap-northeast-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"ap-northeast-2"}],{"Fn::Equals":{"Ref":"AWS:Region"},"ap-south-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"ap-southeast-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"ap-southeast-2"}],{"Fn::Equals":{"Ref":"AWS:Region"},"ca-central-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"cn-north-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"cn-northwest-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"eu-central-1"}],{"Fn::Or":[{"Fn::Equals":{"Ref":"AWS:Region"},"eu-north-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"eu-west-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"eu-west-2"}],{"Fn::Equals":{"Ref":"AWS:Region"},"me-east-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"us-east-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"us-east-2"}],{"Fn::Equals":{"Ref":"AWS:Region"},"us-west-1"}],{"Fn::Equals":{"Ref":"AWS:Region"},"us-west-2"}]}}]

Resources
[+] AWS::SQS::Queue CdkAppQueue CdkAppQueueF2AGFRC
[+] AWS::SQS::QueuePolicy CdkAppQueuePolicy CdkAppQueuePolicy6B032EBA
[+] AWS::SNS::Subscription CdkAppQueue/CdkAppStackCdkAppTopicA685FD2B CdkAppQueueCdkAppStackCdkAppTopicA685FD2B11CAC019
[+] AWS::SNS::Topic CdkAppTopic CdkAppTopicF480CB71

cdk-app on p master [?] ls @v0.1.0 via v14.2.0 using anish-cdk took 2s
```

AWS CDK CLI

Construct levels

L3+

Purpose-built constructs

Opinionated abstractions

L2

AWS constructs

High-level service constructs

L1

CloudFormation resources

Automatically generated

L3 – constructs.dev

Construct Hub

Getting Started ▾Documentation ▾

Build cloud infrastructure faster

Construct Hub is a registry for open-source cloud development kit (CDK) libraries.

Find Constructs

Supported CDKs

We support CDK for CloudFormation (AWS CDK), which generates CloudFormation templates, CDK for Terraform (CDKtf), which generates Terraform-friendly JSON files, and CDK for Kubernetes (CDK8s), which generates Kubernetes manifests.

**AWS CDK****CDK8s****CDKtf**

Supported programming languages

For each CDK library you can find helpful documentation, such as API references and code samples in Java, .NET, TypeScript and Python. Create cloud resources and define your application using loops, conditionals, GitOps, code completion, and manifest testing.

**TypeScript****Python****Java****.NET**

Recently updated

**cdk-simple-serverless**

CDK Constructs for Serverless Demo

**Community**

Version 0.0.3
Published Nov 08, 2021
Author [Pahud Hsieh](#)



**@cdk-cloudformation/awsqs-kubernetes-resource**

Applies a YAML manifest to the specified Kubernetes cluster

**Community**

Version 1.15.0-alpha.1
Published Nov 08, 2021
Author [Amazon Web Services](#)



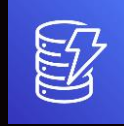


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Custom construct libraries

AWS

L2 constructs



Amazon DynamoDB



Amazon RDS



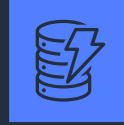
Amazon EC2



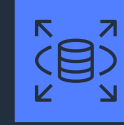
AWS Lambda

L3 constructs

Organization defaults



Your DynamoDB



Your RDS

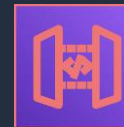
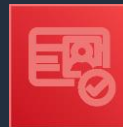


Your EC2



Your Lambda

Common use cases



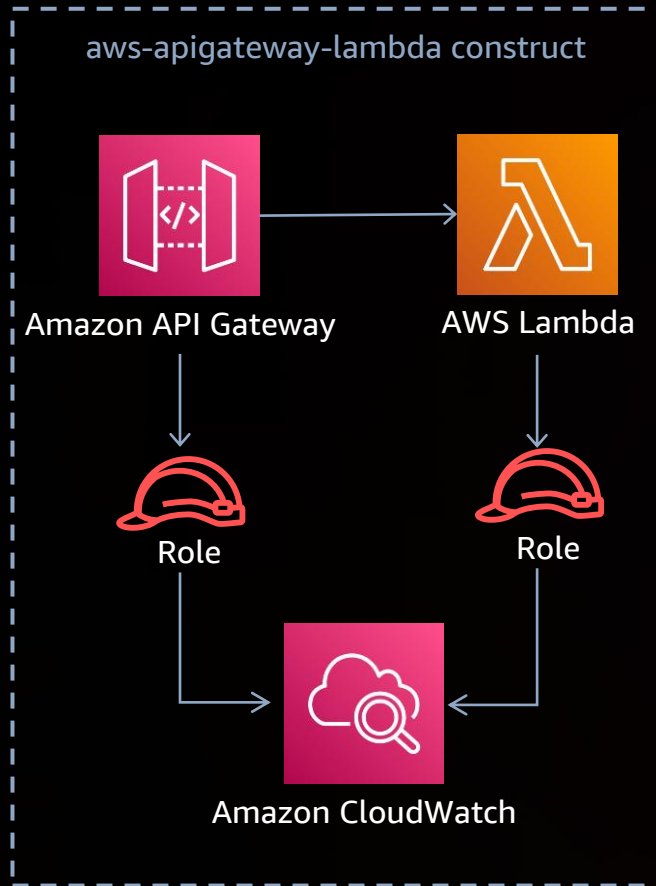
Secure REST API



CloudWatch to
Amazon Kinesis

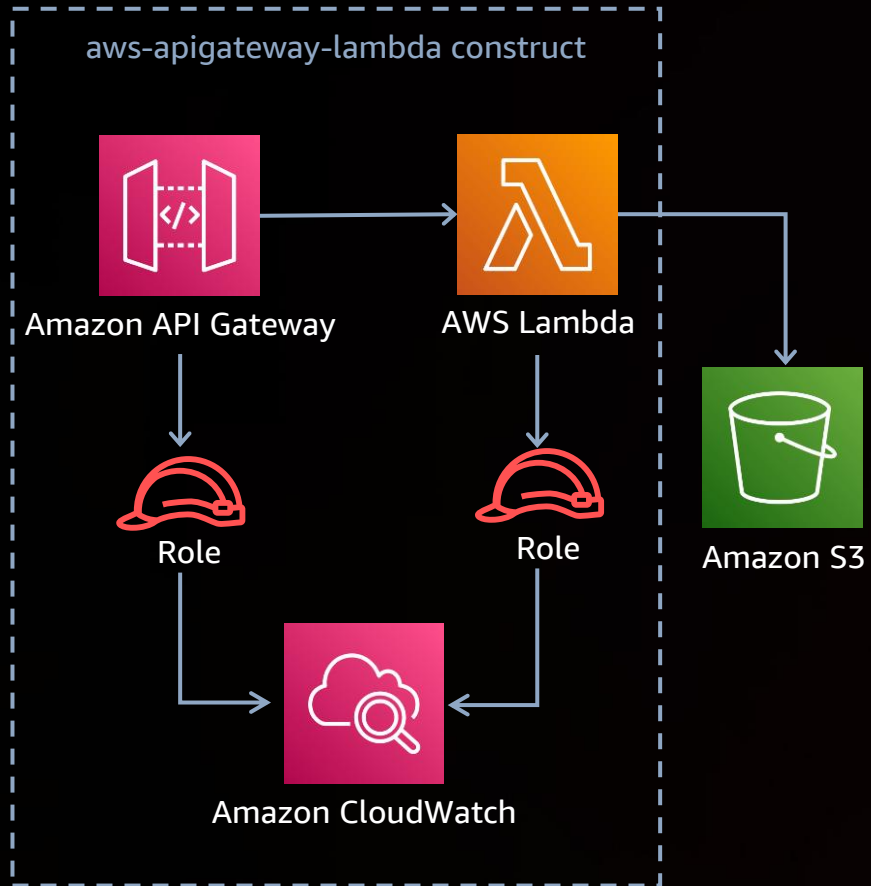
Your organization

Larger building blocks



```
new ApiGatewayToLambda(this, "myApi", {
  lambdaFunctionProps: {
    runtime: lambda.Runtime.NODEJS_14_X,
    code: lambda.Code.fromAsset(`${__dirname}/lambda`),
    handler: "index.handler",
  },
});
```

Composable



```
new ApiGatewayToLambda(this, "myApi", {
  lambdaFunctionProps: {
    runtime: lambda.Runtime.NODEJS_14_X,
    code: lambda.Code.fromAsset(`${__dirname}/lambda`),
    handler: "index.handler",
  },
});

const myBucket = new s3.Bucket(this, "myBucket");

myBucket.grantRead(myApi.lambdaFunction);
```

Well-Architected

- Good defaults
- Integrate services
- Least privilege

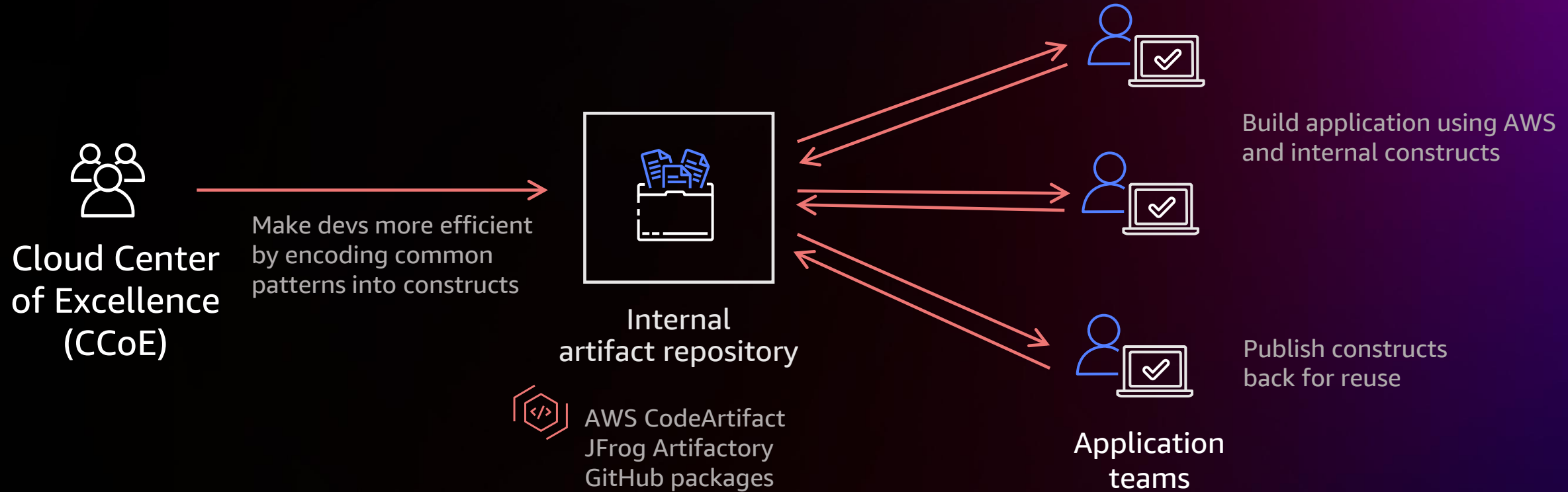
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "xray:PutTelemetryRecords",
        "xray:PutTraceSegments"
      ],
      "Resource": "*",
      "Effect": "Allow"
    },
    {
      "Action": [
        "s3:GetBucket*",
        "s3:GetObject*",
        "s3:List*"
      ],
      "Resource": [
        "arn:aws:s3:::mystack-mybucket5af9c99b",
        "arn:aws:s3:::mystack-mybucket5af9c99b/*"
      ],
      "Effect": "Allow"
    }
  ]
}
```

Testable

- Fine-grained assertions
- Snapshot

```
test('Test properties', () => {  
  // Initial Setup  
  const stack = new Stack();  
  
  const app = new ApiGatewayToLambda(this, "myApi", {  
    lambdaFunctionProps: {  
      runtime: lambda.Runtime.NODEJS_14_X,  
      code: lambda.Code.fromAsset(`${__dirname}/lambda`),  
      handler: "index.handler",  
    },  
  });  
  
  expect(app.lambdaFunction !== null);  
  expect(app.apiGateway !== null);  
  expect(app.apiGatewayCloudWatchRole !== null);  
  expect(app.apiGatewayLogGroup !== null);  
});
```

Sharing custom constructs



Construct anti-patterns

- Security or compliance enforcement
- Tagging enforcement
- Overfit to a specific workload



AWS Security Hub



AWS Config



Amazon GuardDuty



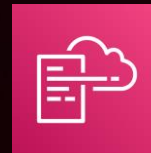
AWS Organizations



Tag policies

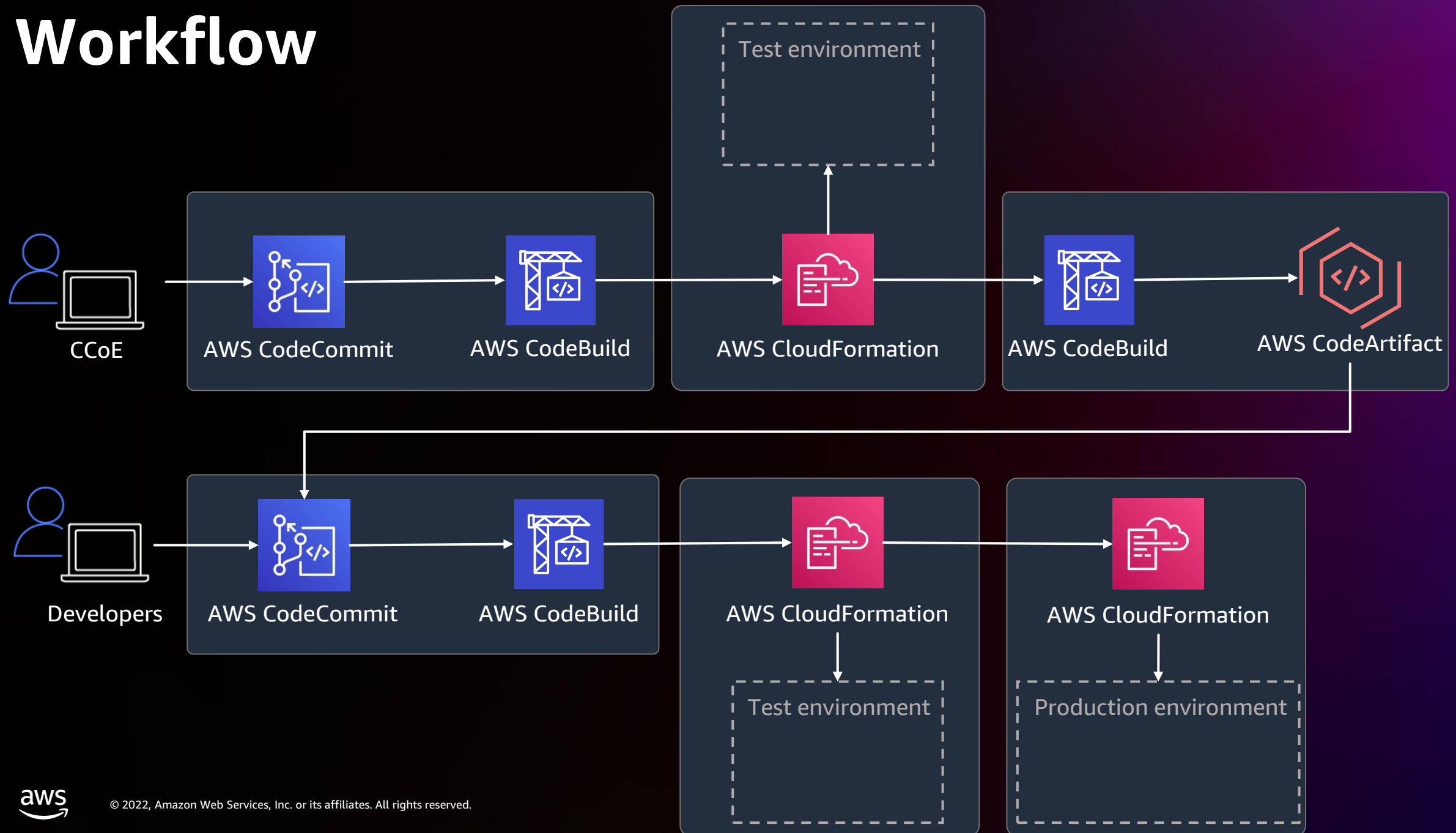


Service control
policies



CloudFormation
hooks

Workflow



Thank you!



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